

Lecture 2 Introduction to Linux Notes

What is an Operating System?

An operating system is the set of instructions that tell the computer how to run. It enables you to use the computers hardware, providing the basic tools that make the computer useful.

What is a Kernel?

A kernel is the core of an operating system. It is a software component that is responsible for managing low-level features of the computer, such as:

- System Hardware
- Memory Allocation
- CPU Time
- Program to Program Interaction

Which other parts aside from the kernel identify an OS?

- Applications
- Graphical Desktop Environment
- Daemons and Shells
- Hardware

What is Linux?

Linux is an open-source, Unix-like operating system that offers many different distributions that are highly scalable and customizable for various implementations.

What is a Linus Distribution?

Linux Distributions are variations and versions of Linux, each one different in their own way, but are all built around the same Linux Kernel.

List at least 4 Linux characteristics:

- Open-Source Software
- Free of Charge
- Highly Scalable and Customizable
- The vast majority of server applications on the internet run on some version of Linux

What is Debian?

Debian is a popular and "feature Complete" distribution of Linux.

List and define the different types of licensing agreements:

- Open-Source: The software may be distributed for a fee or for free, and is distributed with the source code.
- Closed Source: The software is not distributed with the source code, and the user is restricted from modifying the code.
- Free Software: The Software is distributed with the source code.

What is Free Software? Define the 4 freedoms:

Free Software is software distributed for free or with a fee, and is critical to the open source world model.

- Freedom 0: Use the software for any purpose.
- Freedom 1: Examine the source code and modify it as you see fit.
- Freedom 2: Redistribute the software.
- Freedom 3: Redistribute your modified software.

What is virtualization?

Virtualization is defined as creating virtual versions of something. In our case, it is most often used to let multiple Operating Systems run on one physical machine at the same time, allowing administrators to divide hardware and create multiple computers inside one physical machine.